California Polytechnic State University Pomona
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Intro to Microelectronics Laboratory

ECE 2200L
Report #4

LAB 4 – I-V characteristics of Zener diode

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Presented to
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Introduction:

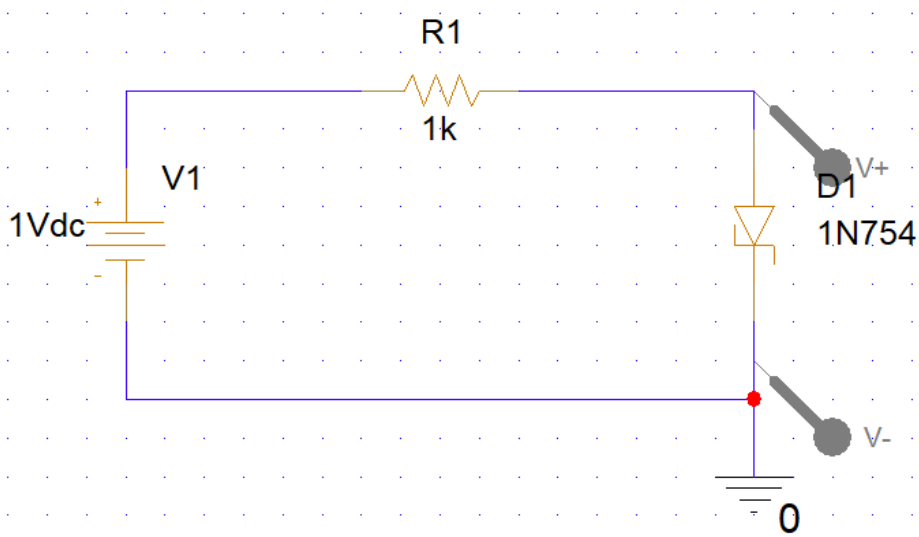
Our objective during this lab is to study the current-voltage relationship of a Zener diode. We will also be determining the saturation current, the ideality factor (η) in the forward region, the Zener voltage, and the Zener resistance in the Zener region. It is important to note that we are using a 5.1V Zener diode.

Parts List:

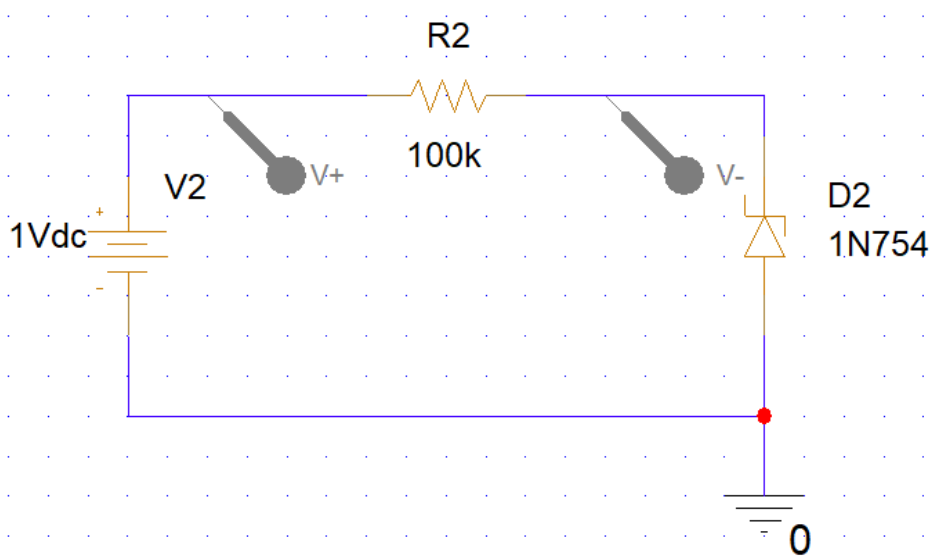
1. Resistors: 220 [Ω], 1 [kΩ], 100 [kΩ]
2. Diodes: 1N751A 5.1 V Zener diode

Circuit Schematics:

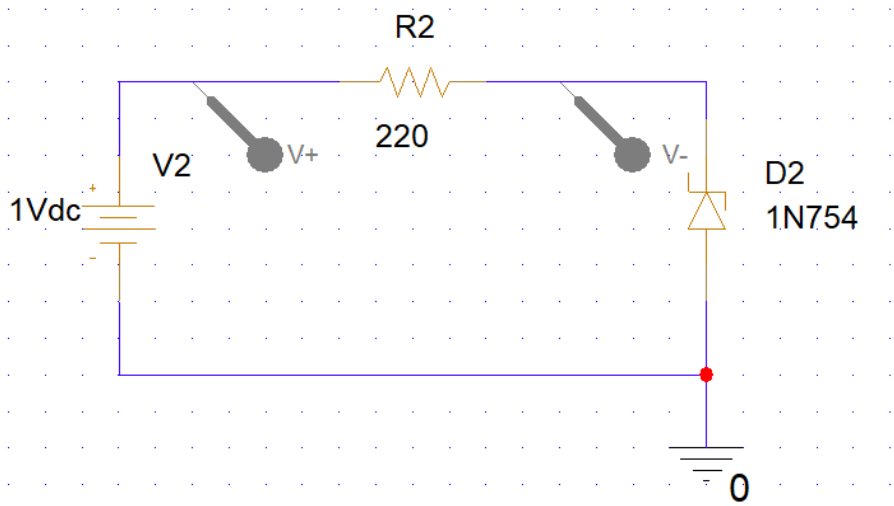
Part 1:



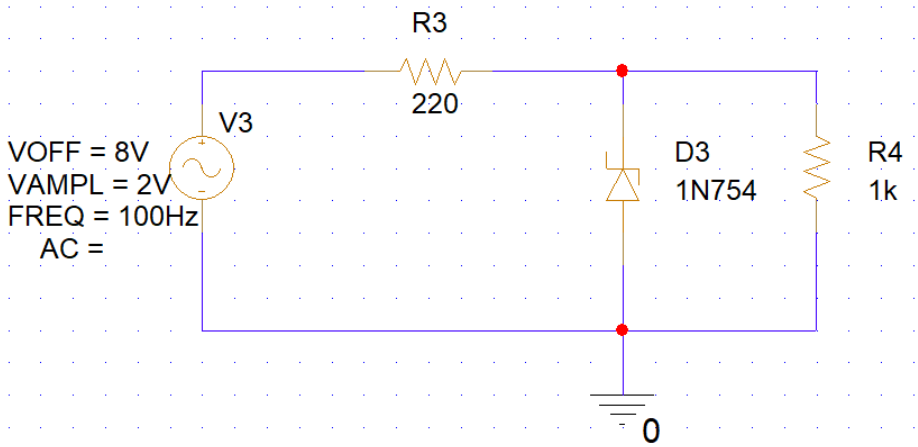
Part 2:



Part 3:



Part 4:



<u>Experimental Data:</u>

Part 1:

Vs (V)	0.1	0.3	0.6	1	2	4	8	12
Vd(V)	0.120	0.317	0.583	0.693	0.734	0.759	0.780	0.786
Vr = Vs-Vd (V)	-0.02 Or 0	-0.017 Or 0	0.017	0.307	1.266	3.241	7.22	11.214
Id=Vr/1 k (mA)	0	0	0.017	0.307	1.266	3.241	7.22	11.214

TABLE 1. Displays data gathered during part 1.

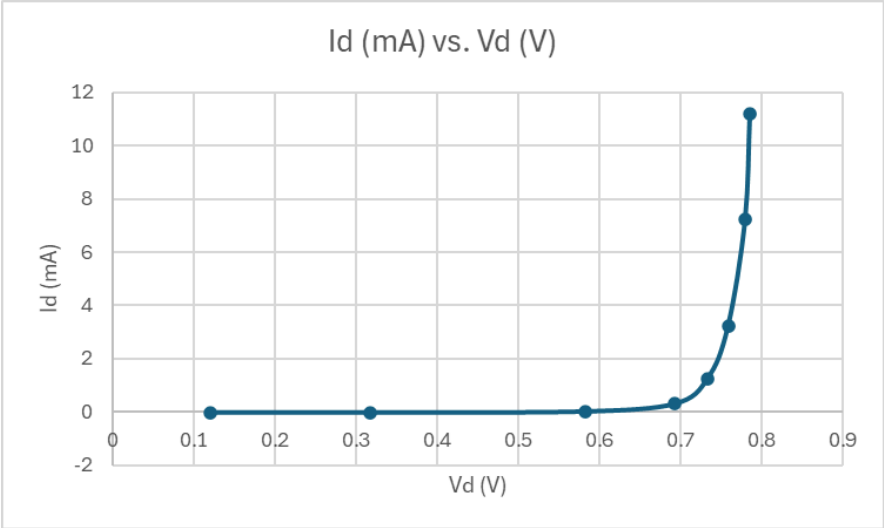


Fig. 1. Depicts Id vs. Vd based on data from Table 1.

Part 2:

Vs (V)	0.5	1	2	3	4
Vr (V)	0	0	0.016	0.257	0.836
Id=Vr/100k (uA)	0	0	0.16	2.57	8.36
Vd=Vs-Vr (V)	0.5	1	1.984	2.743	3.164

TABLE 2. Displays data gathered during part 2.

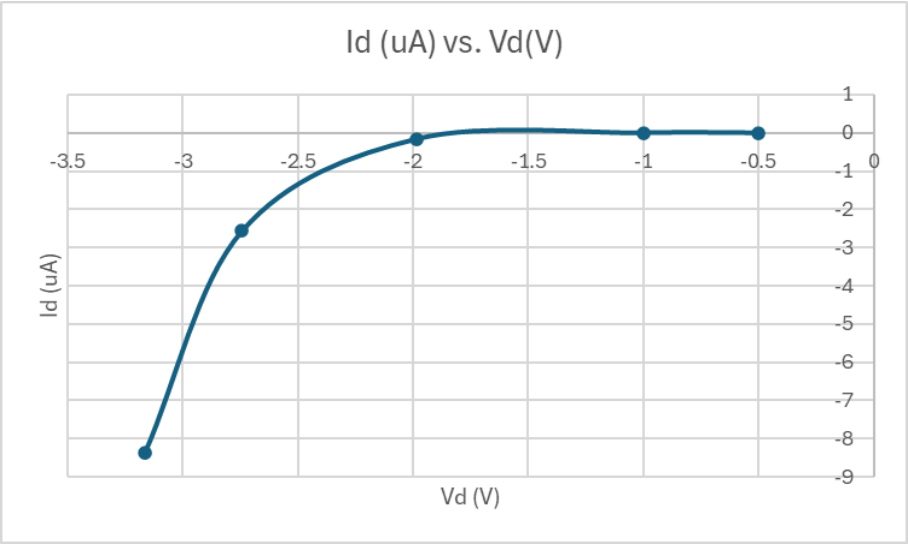


Fig. 2. Depicts Id vs. Vd based on data from Table 2.

Part 3:

Vs (V)	7	8	9	11	13	15
Vd (V)	5.08	5.1	5.13	5.17	5.20	5.24
Id=Vs-Vd/220 (mA)	8.73	13.18	17.6	26.5	35.45	44.36

TABLE 3. Displays data gathered during part 3.

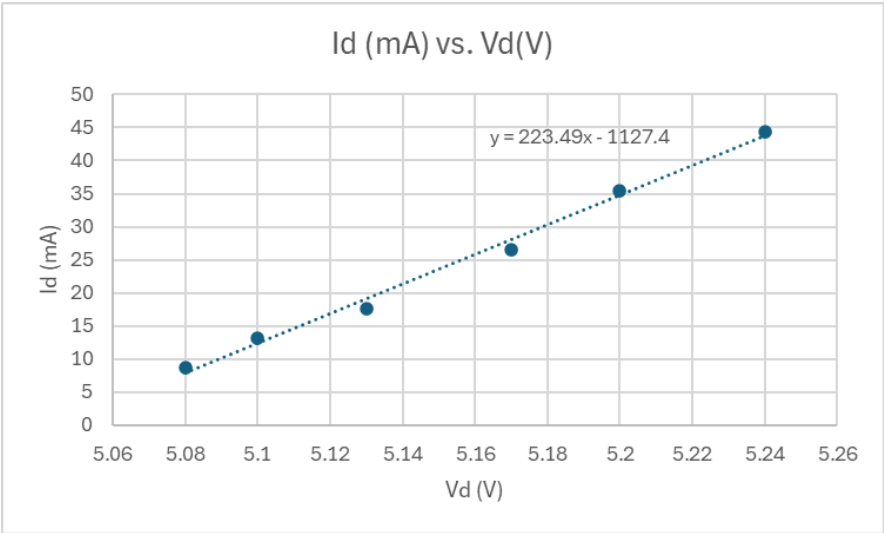


Fig. 3. Depicts Id vs. Vd based on data from Table 3.

Part 4:

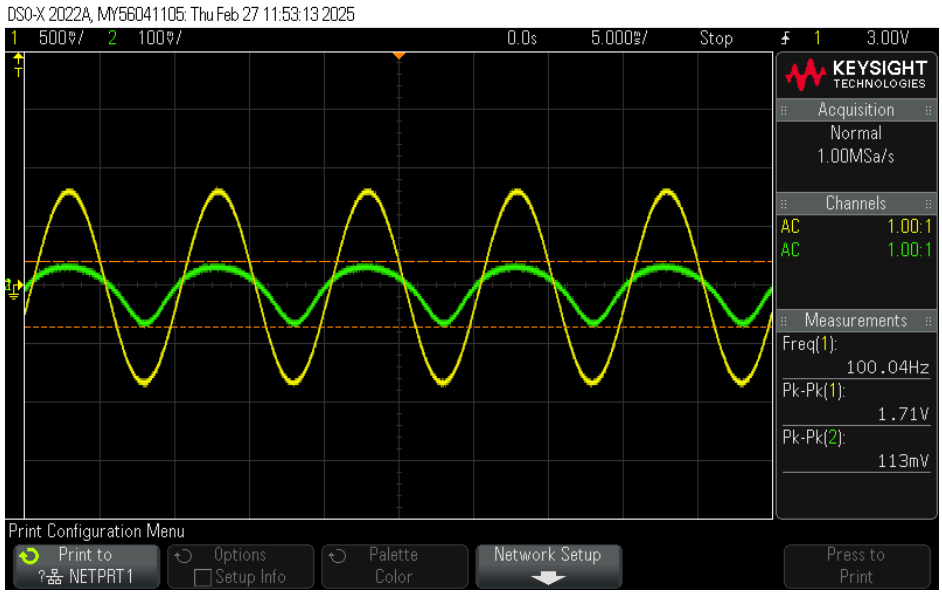
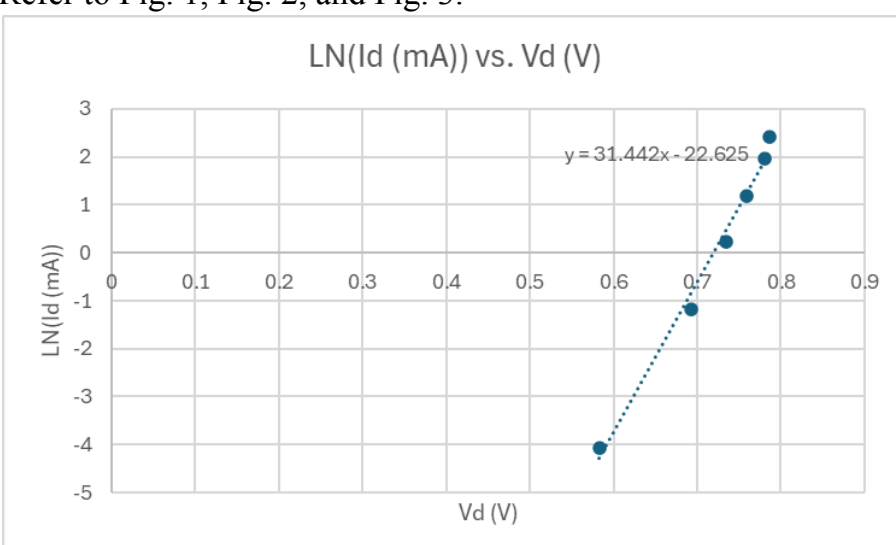


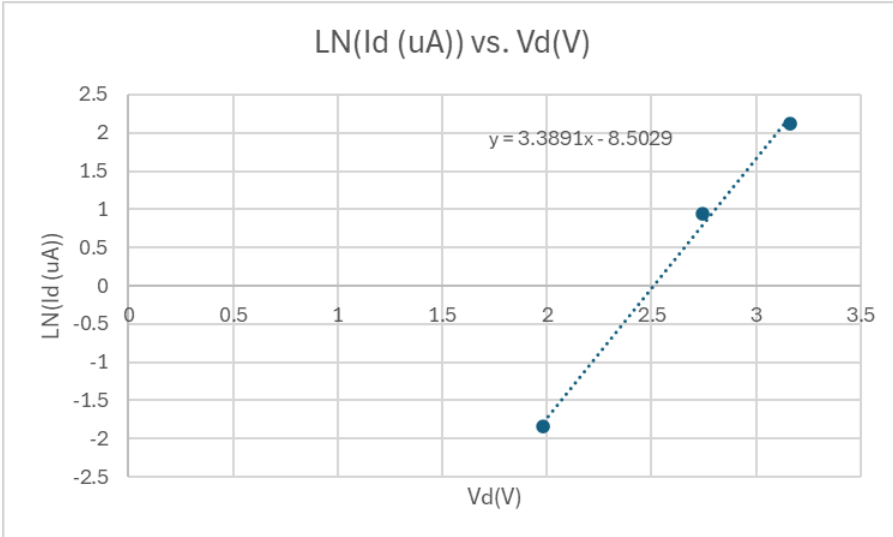
Fig. 3. Screenshot of Vin and Vo signals from part 4 of lab.

Post-Lab:

1. Refer to Fig. 1, Fig. 2, and Fig. 3.



2.



a. η for p1: η = 1/31.442 = 0.0318
b. η for p2: η = 1/3.3891 = 0.2951

3. Reverse saturation current is about 0.16 [μA]
4. Rz = 1/223.49 = 0.004475 [kΩ] = 4.475 [Ω]
5. RRR = 20 * log(Vo / Vs) = -23.598 [dB]

Conclusion:

This lab gave us the opportunity to witness firsthand the behavior of a Zener diode in forward bias, reverse bias, and breakdown. The graphs from part 1 and part 2 clearly illustrate the forward and reverse bias behavior of the diode. The diode experiences an exponential relationship between voltage and current in forward bias, and in reverse it reaches a very small constant value before beginning to exponentially decrease. Our results from part 3 also illustrate the expected behavior of the diode in reverse breakdown. Over very small steps around 5.1 [V], the current increases significantly, showing that the diode has reached breakdown and is allowing current to flow. In part 4 of the lab, we wanted to demonstrate the Zener diode's ability to eliminate ripple voltage. Our Ripple Rejection Ration is a measure of the diode's ability to do this, and our value of -23.598 [dB] is within the range that we expected. The results of part 1, part 2, and part 3 follow what we expected. Though in part 2, our value of 0.16 [μA] is lower than the expected 1 [μA]. Had we taken measurements between 2 [V] and 3 [V], we probably would have found a more accurate saturation current value. Regarding part 3, our Zener resistance of 4.475 [Ω] is within the expected 2 to 10 ohm range. All in all, it seems we successfully verified the diode's behavior, and consequently our understanding of Zener diode behavior has been enhanced. In addition to the theoretical knowledge that we gained during lecture, we now have practical knowledge of how a Zener diode behaves.